Prime-Shell Longitudinal Ledger and the Exact P Collapse

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Abstract

TPC-218 retained the prime label in a Hilbert-valued finite-window estimate, but scalar recovery was recorded only as a factor $P = \#\mathcal{Q}_x$. This paper identifies the factor exactly. For the packet vector $Z_q(n) = (K_{j,q}(n))_j$, we decompose each prime-labelled family into its constant longitudinal mode and its mean-zero transverse remainder. The resulting identity is

$$E_{\text{shell}} = P(E_{\text{diag}} - E_\perp),$$

where $E_\perp$ is the integrated transverse energy. Consequently, an improvement $E_{\text{shell}} \le \eta P E_{\text{diag}}$ is equivalent to the lower bound $E_\perp \ge (1-\eta)E_{\text{diag}}$. Exact rational fixtures attain both the aligned and balanced endpoints. The result is structural: it supplies no prime or Möbius cancellation and no twin-prime conclusion.

Claim level. `PROVED_STRUCTURAL_L1 / EXACT_LONGITUDINAL_TRANSVERSE_LEDGER`. The theorem is an exact Hilbert identity; the finite fixtures are controls, not asymptotic evidence.

1 Why the P factor must be opened

The previous finite-window stages control a common-source kernel after its rational frequencies have been grouped. TPC-218 kept (q, j) as coordinates and proved a split bound at the scale $Q^2/H = x^{1/96}$, but the packet shell $K_j = \sum_{q\in\mathcal{Q}_x} K_{j,q}$ was recovered by pointwise Cauchy. That estimate is safe, but it hides the precise object on which a future signed argument must operate. The inherited finite-window estimate uses the standard additive large-sieve interface [1]; the present paper audits what happens after its q labels are recombined.

The present paper asks a narrower question: for an arbitrary family of packet vectors, what is the exact difference between the q-diagonal energy and the scalar shell energy? This question is deliberately independent of any conjectural prime distribution. It is an algebraic audit of the interface that a prime-shell theorem would have to cross.

2 The labelled packet object

Let $\mathcal{Q}_x$ be a nonempty finite set of primes and put $P = \#\mathcal{Q}_x$. Let $V = \mathbb{C}^J$ with its standard Hermitian norm. In the TPC-218 application,

$$Z_q(n) = \big(K_{j,q}(n)\big)_{0\le j<J} \in V, \qquad K_j(n) = \sum_{q\in\mathcal{Q}_x} K_{j,q}(n).$$

Nothing in the next theorem uses the formula for $K_{j,q}$; it only uses that all terms belong to one common Hilbert space and share the same interval.

For a finite interval I define

$$\overline{Z}(n) = \frac{1}{P} \sum_{q \in \mathcal{Q}_x} Z_q(n), \qquad R_q(n) = Z_q(n) - \overline{Z}(n),$$

and

$$\begin{aligned} E_{\text{shell}} &= \sum_{n \in I} \left\| \sum_{q \in \mathcal{Q}_x} Z_q(n) \right\|_2^2, \\ E_{\text{diag}} &= \sum_{n \in I} \sum_{q \in \mathcal{Q}_x} \|Z_q(n)\|_2^2, \\ E_{\perp} &= \sum_{n \in I} \sum_{q \in \mathcal{Q}_x} \|R_q(n)\|_2^2. \end{aligned}$$

The names longitudinal and transverse refer only to the decomposition of the direct sum V^P into the constant q-direction and its orthogonal complement. They do not assume that the primes behave randomly.

3 Exact longitudinal/transverse theorem

Theorem 1 (Exact P-collapse ledger)**.** *For every family $(Z_q(n))$ as above,*

$$E_{\text{shell}} = P(E_{\text{diag}} - E_{\perp}), \qquad 0 \le E_{\perp} \le E_{\text{diag}},$$

and hence $0 \le E_{\text{shell}} \le P E_{\text{diag}}$. For $0 \le \eta \le 1$,

$$E_{\text{shell}} \le \eta P E_{\text{diag}} \quad \iff \quad E_{\perp} \ge (1-\eta) E_{\text{diag}}.$$

Proof. For each n, the residuals satisfy $\sum_q R_q(n) = 0$. Expanding the squared norms, the cross term with $\overline{Z}(n)$ therefore vanishes:

$$\sum_q \|Z_q(n)\|_2^2 = P\|\overline{Z}(n)\|_2^2 + \sum_q \|R_q(n)\|_2^2. \tag{1}$$

On the other hand, $\sum_q Z_q(n) = P\overline{Z}(n)$, so

$$\left\| \sum_q Z_q(n) \right\|_2^2 = P^2 \|\overline{Z}(n)\|_2^2.$$

Eliminating the mean term with (1) gives the pointwise identity

$$\left\| \sum_q Z_q(n) \right\|_2^2 = P \sum_q \|Z_q(n)\|_2^2 - P \sum_q \|R_q(n)\|_2^2. \tag{2}$$

Summing (2) over $n \in I$ proves the first equality. Equation (1) and nonnegativity give $0 \le E_{\perp} \le E_{\text{diag}}$. Finally, rearranging the first equality proves the last equivalence; every operation divides only by the positive integer P. □

Corollary 2 (What a genuine shell saving must prove)**.** *If a future argument claims a factor $\eta < 1$ relative to the TPC-218 Cauchy envelope, then it must prove a lower bound on the literal q-transverse energy of the form $E_{\perp} \ge (1-\eta)E_{\text{diag}}$ on the same interval and with the same source object. An upper bound for E_{diag} alone cannot imply this claim.*

fixture	E_{diag} / $E_{\perp}$	E_{shell}
aligned	20/0	$80 = PE_{\mathrm{diag}}$
balanced	4/4	0
orthogonal	4/4	0
mixed	10/8	8

Table 1: Exact rational endpoint and interior checks with $P = 4$.

4 Sharp finite endpoints

The theorem has no slack at the level of abstract geometry. If $Z_q = v$ for every q, then $R_q = 0$, so $E_{\perp} = 0$ and $E_{\mathrm{shell}} = PE_{\mathrm{diag}}$. At the other endpoint, if $\sum_q Z_q = 0$, then $\overline{Z} = 0$, $E_{\perp} = E_{\mathrm{diag}}$, and $E_{\mathrm{shell}} = 0$.

Table 1 records the exact finite certificate. The vectors are rational, so no floating-point decision is used. The aligned row is the same structural pattern as the TPC-218 $d = 5$ fixture, but the present theorem explains it as a zero-transverse endpoint rather than merely a large ratio.

The aligned fixture refutes a geometry-only hope for a sub-P estimate. The statement is scoped: it does not say that the literal growing prime shell is aligned. It says that such alignment cannot be excluded by the Hilbert envelope or by the label space alone.

5 Route position and reproducibility

Route A is not applicable. Route B structural threshold A passes because the theorem acts on the exact labelled interface inherited from TPC-218. The maximum claim is `PROVED_STRUCTURAL_L1`. In particular,

$$\texttt{ARITHMETIC\_ADVANCE} = \texttt{NO}, \quad \texttt{FIXED\_ATOM\_CREDIT} = 0, \quad \texttt{L2} = \texttt{NONE},$$

and the strict 1/400 payment remains unpaid.

The producer, an independent checker, an optimized-mode replay, and a separate endpoint adversary are included in the project directory. The next natural edge is no longer ambiguous: rewrite $E_{\perp}$ using the congruence collisions of the literal prime rows. That is the TPC-220 question.

6 Conclusion

TPC-219 converts the scalar P collapse from an opaque Cauchy cost into an exact longitudinal/transverse ledger. Any successful signed prime-shell reassembly must create transverse energy in the literal q family; the aligned endpoint shows why this cannot be obtained from abstract packet geometry. The result is a structural bridge, not an arithmetic saving.

References

[1] Hugh L. Montgomery and Robert C. Vaughan. The large sieve. *Mathematika*, 20:119–134, 1973.